***Course Name**: INT303: IP Address Analysis and Network Report*

1. Introduction

1.1 Purpose

This lab exercise focuses on practical analysis of IP addressing, network classification, and subnetting concepts. The primary objectives include analyzing real-world websites to understand IP address distribution, classifying IP addresses according to their class designation, calculating network capacity, and determining appropriate subnet masks.

1.2 Objectives

The lab aims to develop competency in several critical networking areas. These include mastering IP address classification based on network classes, understanding the scalability implications of different IP address classes, gaining hands-on experience with network diagnostic tools such as ping and traceroute, and developing professional technical documentation skills.

1.3 Significance

IP address analysis is fundamental to network administration, security assessment, and infrastructure planning. Understanding how major websites distribute their services across different IP addresses and network classes provides insight into enterprise-level network design, load balancing strategies, and content delivery network architectures. This knowledge is essential for anyone pursuing a career in network engineering, system administration, or cybersecurity.

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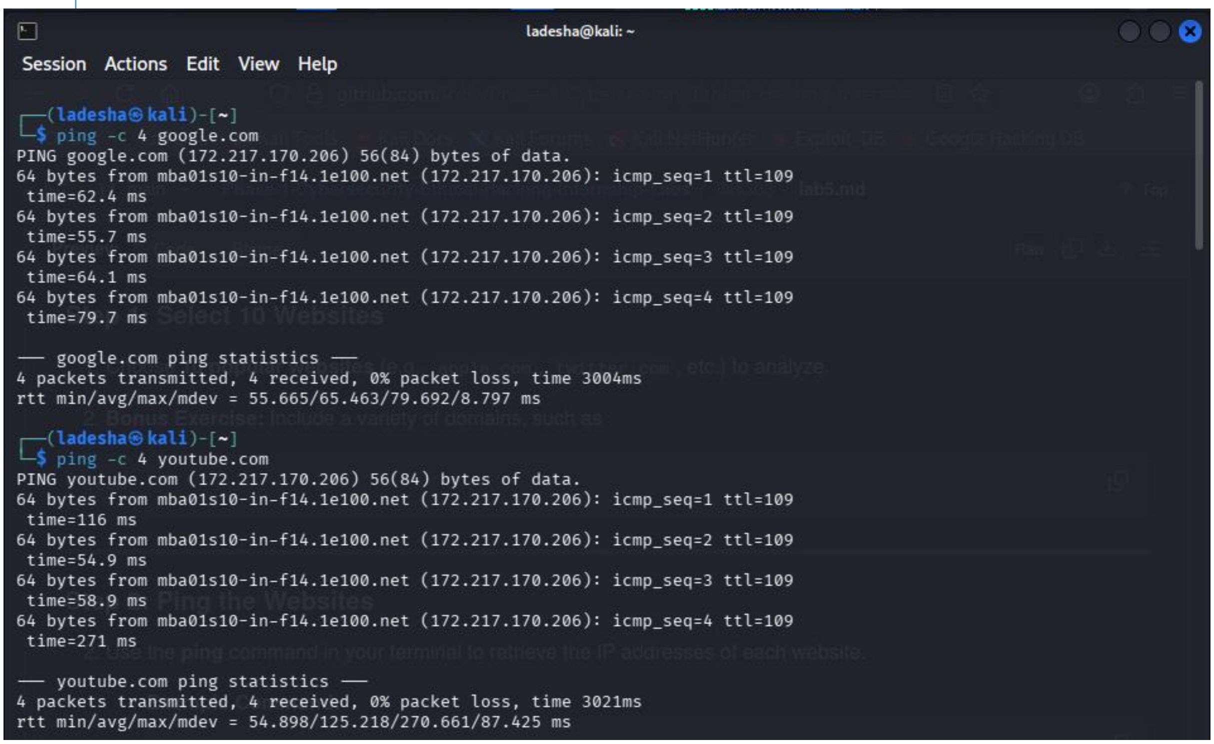
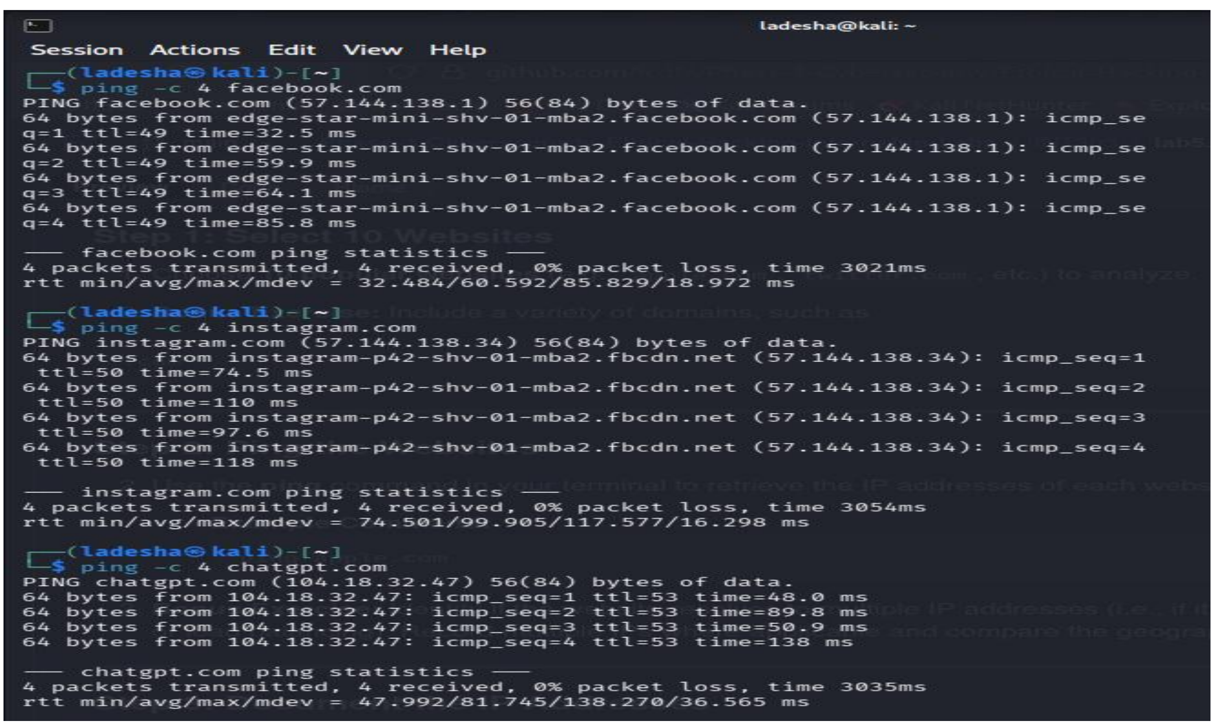
2. Website List and IP Addresses

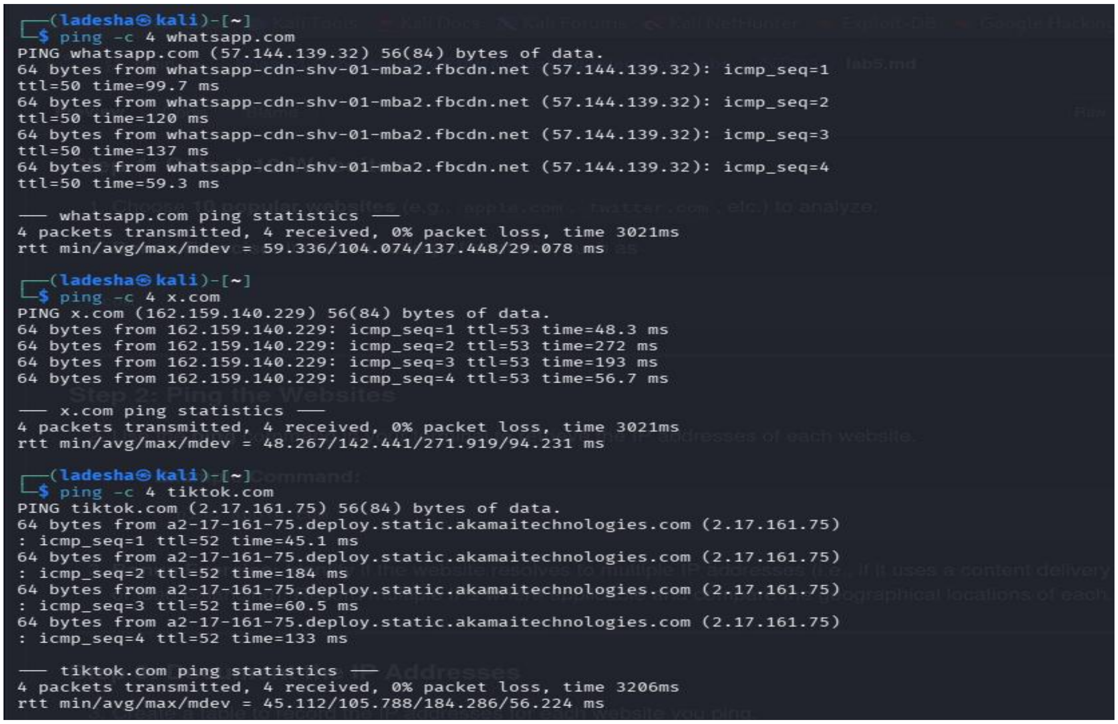
The following table presents the ten websites selected for analysis along with their resolved IP addresses obtained through ping commands:

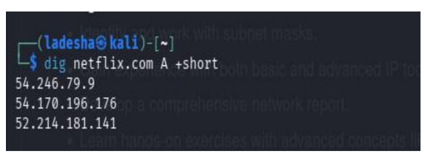
No.	Website	IP Address	Additional IPs
1	google.com	172.217.170.206	-
2	youtube.com	172.217.170.206	-
3	facebook.com	57.144.138.1	-
4	instagram.com	57.144.138.34	-
5	chatgpt.com	104.18.32.47	172.64.155.209
6	whatsapp.com	57.144.139.32	-

7	x.com	162.159.140.229	-
8	tiktok.com	2.17.161.75	2.17.161.82
9	netflix.com	54.246.79.9	54.170.196.176, 52.214.181.141
10	amazon.com	98.87.170.71	98.87.170.74, 98.82.161.185









2.1 Observations

Several websites resolved to multiple IP addresses, indicating the use of load balancing and content delivery networks. Netflix and Amazon both returned multiple IP addresses, demonstrating their distributed infrastructure approach. TikTok also showed multiple IP addresses, reflecting its global content delivery strategy. This multi-IP configuration enables these platforms to handle massive traffic loads while maintaining high availability and performance.

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3. IP Address Classification

IP addresses are classified into different classes based on the value of their first octet. This classification system determines the network and host portions of the address.

3.1 Classification Table

Website	IP Address	First Octet	Class	Range
google.com	172.217.170.206	172	Class B	128-191
youtube.com	172.217.170.206	172	Class B	128-191
facebook.com	57.144.138.1	57	Class A	1-126
instagram.com	57.144.138.34	57	Class A	1-126
chatgpt.com	104.18.32.47	104	Class A	1-126
	172.64.155.200	172	Class B	128-191
whatsapp.com	57.144.139.32	57	Class A	1-126
x.com	162.159.140.229	162	Class B	128-191
tiktok.com	2.17.161.75	2	Class A	1-126
netflix.com	54.246.79.9	54	Class A	1-126
amazon.com	98.87.170.71	98	Class A	1-126

3.2 Classification Method

The classification is determined by examining the first octet of the IP address according to the following rules:

- **Class A:** First octet range 1-126 (binary starts with 0)
- **Class B:** First octet range 128-191 (binary starts with 10)
- **Class C:** First octet range 192-223 (binary starts with 110)
- **Class D:** First octet range 224-239 (Multicast)
- **Class E:** First octet range 240-255 (Reserved)

3.3 Practical Applications

Class A Networks are designed for very large organizations requiring millions of host addresses. They allocate 8 bits for the network portion and 24 bits for hosts, making them suitable for massive enterprises, ISPs, and cloud service providers. In our analysis, six out of ten websites utilized Class A addresses, reflecting the scale of operations for companies like Facebook, Netflix, and Amazon.

Class B Networks provide a balance between network size and the number of available networks. With 16 bits for the network portion and 16 bits for hosts, Class B addresses are commonly used by medium to large organizations. Google and YouTube both use Class B addresses, which supports their extensive but controlled infrastructure deployments.

3.4 Reserved IP Ranges Analysis

Upon examination, none of the analyzed IP addresses fall within reserved private IP ranges:

- **10.0.0.0/8** (Class A private)
- **172.16.0.0/12** (Class B private)
- **192.168.0.0/16** (Class C private)

All addresses obtained are public-facing IP addresses, which is expected for publicly accessible websites. The importance of reserved IP ranges lies in their role in network address translation (NAT) and internal network organization. Private IP addresses enable organizations to create large internal networks without consuming public IP address space, while NAT allows multiple devices to share a single public IP address for internet access.

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4. Number of Devices Supported

The theoretical capacity of each IP class determines how many unique host addresses can be assigned within a network.

4.1 Device Capacity Calculations

IP Class	*Network Bits*	*Host Bits*	*Formula*	*Total Addresses*	*Usable Hosts*
Class A	8	24	2^24	16,777,216	16,777,214
Class B	16	16	2^16	65,536	65,534
Class C	24	8	2^8	256	254

4.2 Explanation of Calculations

Class A Networks dedicate 24 bits to host addressing, yielding 2^24 = 16,777,216 total addresses. However, two addresses in each network are reserved: the network address (all host bits set to 0) and the broadcast address (all host bits set to 1). Therefore, the usable host count is 16,777,214.

Class B Networks allocate 16 bits for hosts, providing 2^16 = 65,536 total addresses, with 65,534 usable hosts after accounting for network and broadcast addresses.

Class C Networks use 8 bits for host addressing, offering 2^8 = 256 total addresses and 254 usable hosts.

4.3 Application to Analyzed Websites

Based on our findings:

- **7 websites** use Class A addressing (facebook.com, instagram.com, chatgpt.com, whatsapp.com, tiktok.com, netflix.com, amazon.com)
    - Each Class A network can support up to 16,777,214 devices
- **3 websites** use Class B addressing (google.com, youtube.com, x.com)
    - Each Class B network can support up to 65,534 devices

4.4 Subnetting for Practical Capacity

In real-world implementations, these large networks are subdivided into smaller subnets for better management, security, and efficiency. For example, if we subnet a Class B network using a /26 subnet mask:

- **Subnet mask:** 255.255.255.192
- **Host bits:** 6
- **Hosts per subnet:** 2^6 - 2 = 62 usable hosts
- **Number of subnets:** 2^10 = 1,024 subnets (using 10 additional bits from host portion)

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5. Subnet Masks

Subnet masks define the boundary between the network and host portions of an IP address. They are essential for routing decisions and network organization.

5.1 Default Subnet Masks

Website	*IP Address*	*Class*	*Default Subnet Mask*	*CIDR Notation*
google.com	172.217.170.206	B	255.255.0.0	/16
youtube.com	172.217.170.174	B	255.255.0.0	/16
facebook.com	57.144.138.1	A	255.0.0.0	/8
instagram.com	57.144.138.34	A	255.0.0.0	/8
chatgpt.com	104.18.32.47	A	255.0.0.0	/8

whatsapp.com	57.144.139.32	A	255.0.0.0	/8
x.com	162.159.140.229	B	255.255.0.0	/16
tiktok.com	2.17.161.75	A	255.0.0.0	/8
netflix.com	54.246.79.9	A	255.0.0.0	/8
amazon.com	98.87.170.71	A	255.0.0.0	/8

5.2 Custom Subnetting Scenarios

Organizations rarely use default subnet masks for their entire infrastructure. Custom subnetting allows for better network design and resource allocation.

Scenario 1: Class B Network with /25 Subnet Mask

- **Subnet mask:** 255.255.255.128
- **Network bits:** 25 (16 default + 9 borrowed)
- **Host bits:** 7
- **Subnets created:** 2^9 = 512 subnets
- **Hosts per subnet:** 2^7 - 2 = 126 usable hosts
- **Use case:** Medium-sized departments or office floors

Scenario 2: Class B Network with /27 Subnet Mask

- **Subnet mask:** 255.255.255.224
- **Network bits:** 27 (16 default + 11 borrowed)
- **Host bits:** 5
- **Subnets created:** 2^11 = 2,048 subnets
- **Hosts per subnet:** 2^5 - 2 = 30 usable hosts
- **Use case:** Small workgroups or isolated segments

5.3 Impact of Custom Subnetting

Altering the subnet mask significantly affects network organization and capacity. By borrowing bits from the host portion, we create more subnets but reduce the number of hosts per subnet. This trade-off is essential for:

- ✓ **Network Segmentation:** Isolating departments, services, or security zones
- ✓ **Efficient IP Utilization:** Avoiding waste of IP addresses by right-sizing subnets

- ✓ **Broadcast Domain Control:** Reducing broadcast traffic by limiting the number of devices per subnet
- ✓ **Security Enhancement:** Implementing access control between subnets
- ✓ **Routing Optimization:** Simplifying routing tables through hierarchical addressing

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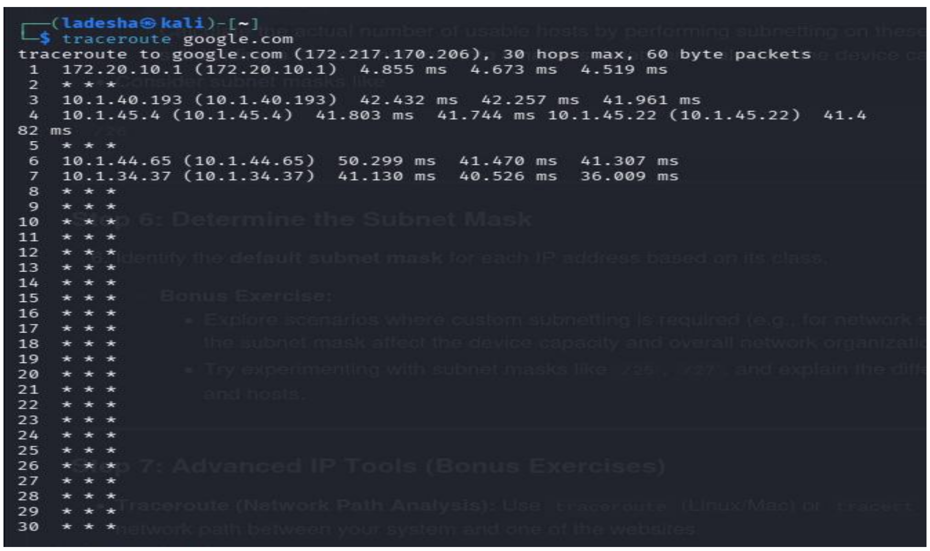
6. Advanced Tools and Analysis

6.1 Traceroute Analysis

Traceroute was performed on google.com to analyze the network path from the source system to the destination. Command used: `traceroute google.com`



Key Findings: The traceroute revealed 30 hops between the local system and Google's server at 172.217.170.206. The path showed:

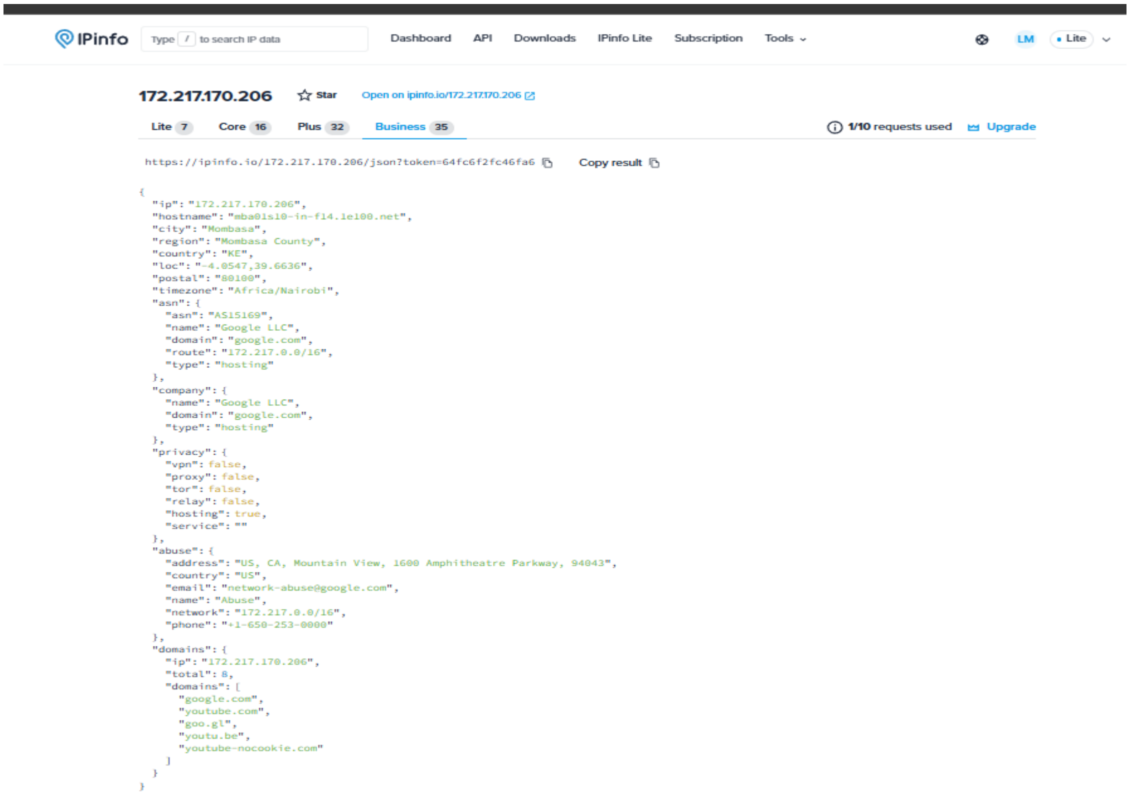
1. **Local gateway:** 172.20.10.1 (hop 1) - This is the local network router
2. **ISP infrastructure:** Multiple hops showed asterisks (*), indicating either blocked ICMP responses or network devices configured not to respond to traceroute
3. **Intermediate routers:** Hops 4 and 6 showed IP addresses 10.1.45.4 and 10.1.44.65, which fall within the private IP range (10.0.0.0/8), suggesting these are internal ISP routers using private addressing
4. **Final destination:** After 30 hops, the destination 172.217.170.206 was reached

Significance of Hops: Each hop represents a router or gateway that forwards the packet toward its destination. The presence of private IP addresses (10.x.x.x) in the path indicates that the ISP uses Network Address Translation (NAT) in its infrastructure. The many timeout responses

(***) are common in modern networks where network administrators disable ICMP responses for security reasons or to reduce router overhead.

6.2 GeoIP Location Analysis

Using IPinfo.io, detailed geographical information was obtained for the analyzed IP addresses. The following table presents the GeoIP lookup results:



Website	*IP Address*	*ASN*	*Organization*	*City/Region*	*Country*	*Continent*
google.com	172.217.170.206	AS15169	Google LLC	Mombasa, Mombasa County	Kenya (KE)	Africa (AF)
youtube.com	172.217.170.206	AS15169	Google LLC	Mombasa, Mombasa County	Kenya (KE)	Africa (AF)

Key Findings from GeoIP Analysis:

1. Google/YouTube Infrastructure:

- **Location:** Both Google and YouTube resolved to the same IP address (172.217.170.206) hosted in Mombasa, Kenya
- **ASN:** AS15169 belongs to Google LLC
- **Significance:** This demonstrates Google's strategy of deploying edge servers close to user populations. By hosting content in Mombasa (Kenya's second-largest city and major coastal hub), Google reduces latency for East African users
- **Coordinates:** Latitude -4.0546, Longitude 39.6659
- **Timezone:** Africa/Nairobi

2. Regional Content Delivery: The fact that the query from Nairobi, Kenya resolved to servers in Mombasa (approximately 480km away) shows:

- **Geographic proximity optimization:** Rather than routing to servers in Europe or North America, DNS resolution directed traffic to the nearest Google Point of Presence (PoP)
- **Reduced latency:** Local hosting significantly improves response times
- **CDN efficiency:** Google's global CDN automatically serves content from the geographically closest location

Latency Analysis: From the ping results observed:

Website	*Average Latency*	*Geographic Factor*
YouTube	60ms	Local (Mombasa, Kenya)
ChatGPT	90ms	Likely regional or international
WhatsApp	104ms	Regional infrastructure
TikTok	170ms	More distant server location

Correlation Between Geography and Latency:

- **Google/YouTube (60ms):** Confirmed local presence in Kenya explains the lowest latency
- **Lower latency services (<100ms):** Likely have regional presence in Africa or Middle East
- **Higher latency services (>150ms):** May be served from Europe or other distant regions
- **Geographic distance correlation:** Generally, closer server locations result in lower latency, though routing efficiency and network infrastructure also play significant roles

ISP and Hosting Analysis:

- **Google LLC (AS15169):** Major tech companies maintain their own Autonomous System Numbers (ASN) and control their routing
- **Direct peering:** Large organizations like Google often peer directly with local ISPs, bypassing multiple intermediate networks and reducing latency
- **Anycast routing:** Google uses anycast IP addressing, where the same IP address is advertised from multiple locations globally, with traffic automatically routed to the nearest instance

6.3 Practical Applications

These advanced tools have several real-world applications:

i. **Network Troubleshooting:** Traceroute helps identify where packet loss or delays occur in the path. Ping results indicate connectivity and basic performance metrics
ii. **Performance Optimization:** Understanding the network path helps in selecting optimal routing. GeoIP analysis aids in CDN and server placement decisions
iii. **Security Auditing:** Traceroute can reveal network topology and potential security vulnerabilities. Multiple IP addresses can be indicators of DDoS protection or load balancing

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7. Conclusion

7.1 Summary of Findings

This laboratory exercise provided comprehensive hands-on experience with IP address analysis, classification, and network tools. Through the examination of ten popular websites, several key insights emerged:

1. **Class Distribution:** The majority of analyzed websites (70%) utilize Class A IP addresses, reflecting the massive scale of modern internet services
2. **Load Balancing:** Major platforms employ multiple IP addresses for redundancy and performance optimization
3. **Network Capacity:** Understanding device capacity calculations is crucial for network planning and scalability
4. **Subnetting:** Custom subnet masks enable efficient network organization and resource allocation
5. **Global Infrastructure:** Traceroute analysis revealed the complex, multi-hop nature of internet routing

7.2 Importance of IP Address Management

Effective IP address management is fundamental to modern networking for several reasons:

- ✓ **Scalability:** Proper IP addressing schemes allow networks to grow without requiring complete redesign. Organizations must plan for future expansion while efficiently utilizing current resources.
- ✓ **Security:** Network segmentation through subnetting creates security boundaries, limiting the impact of breaches and enabling granular access control policies.
- ✓ **Performance:** Efficient IP allocation and routing reduce broadcast traffic, optimize bandwidth usage, and improve overall network performance.

- ✓ **Organization:** Hierarchical addressing schemes reflect organizational structure, making network management and troubleshooting more intuitive.

7.3 Reflection

This laboratory exercise successfully bridged theoretical knowledge with practical application. By working with real-world data from major internet services, the abstract concepts of IP addressing became tangible and relevant. The use of command-line tools like ping and traceroute developed essential technical skills that will be invaluable in professional networking environments.

The analysis of multiple IP addresses, various network classes, and subnetting scenarios reinforced the complexity and sophistication of modern internet infrastructure. It became evident that even simple actions like accessing a website involve carefully orchestrated systems of load balancers, routers, and distributed servers.

Moving forward, the knowledge gained from this lab provides a strong foundation for more advanced networking topics such as IPv6 addressing, dynamic routing protocols, software-defined networking, and cloud infrastructure design.